



Insights from JWST mid-IR instrument about the Galactic Centre

Tamojeet Roychowdhury

UC Berkeley Astronomy

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Timeline of Discovery

1974: Bright radio source discovered (Balick & Brown)

2003: X-ray flare discovered (Chandra, Baganoff et al.)

NIR-submm simultaneous flare (Eckart et al.)

2002-2004: Black hole mass constrained, other sources ruled out (Ghez, Genzel)

2020: EHT images of Sgr A*, disk seen clearly with no jet

2001: X-ray source discovered (Chandra, Baganoff et al.)

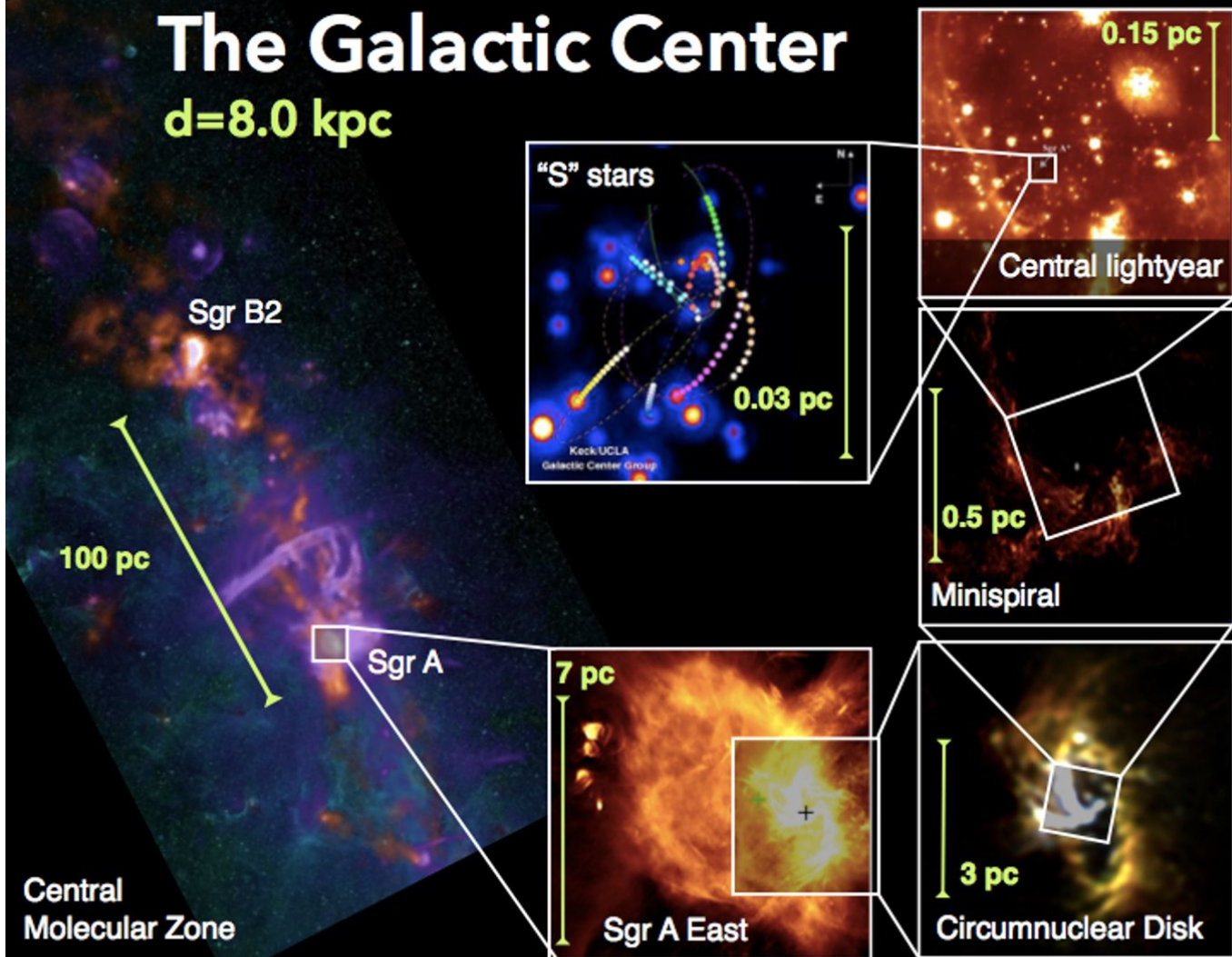
2000-2004: Disk and jet models (synchrotron, SSC, free-free)

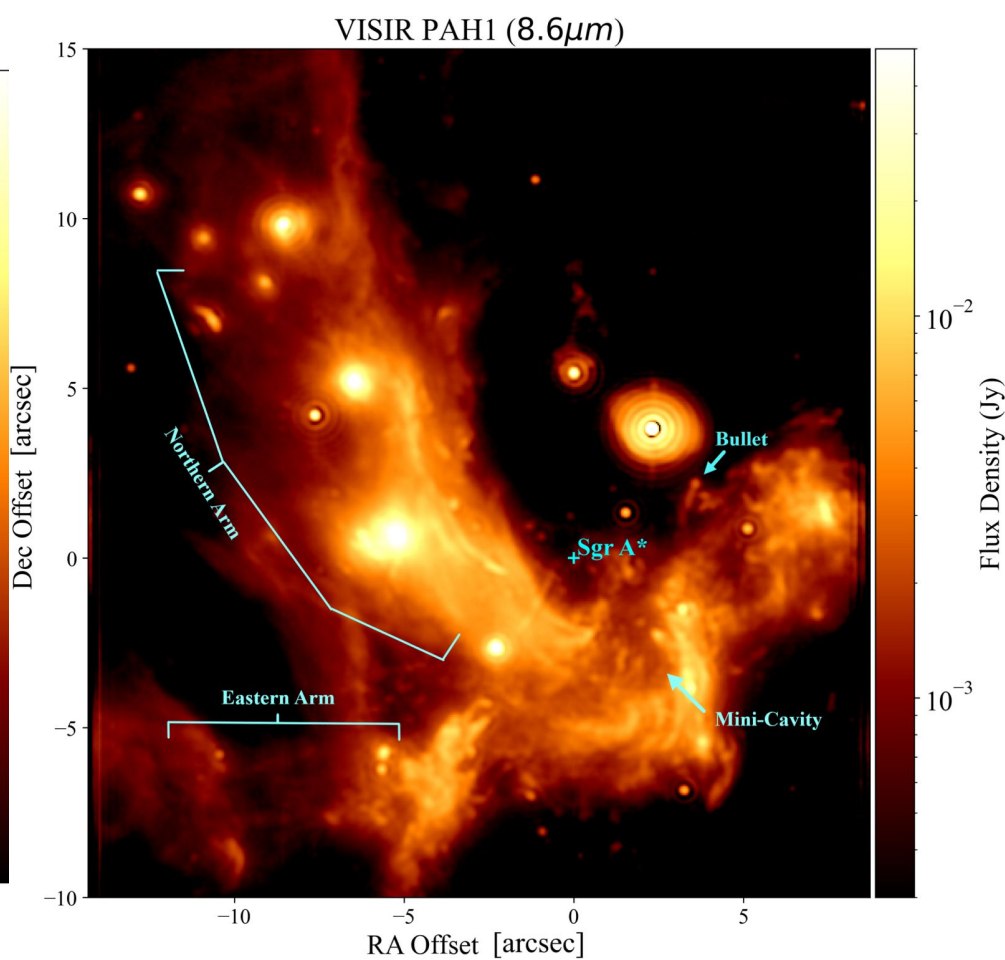
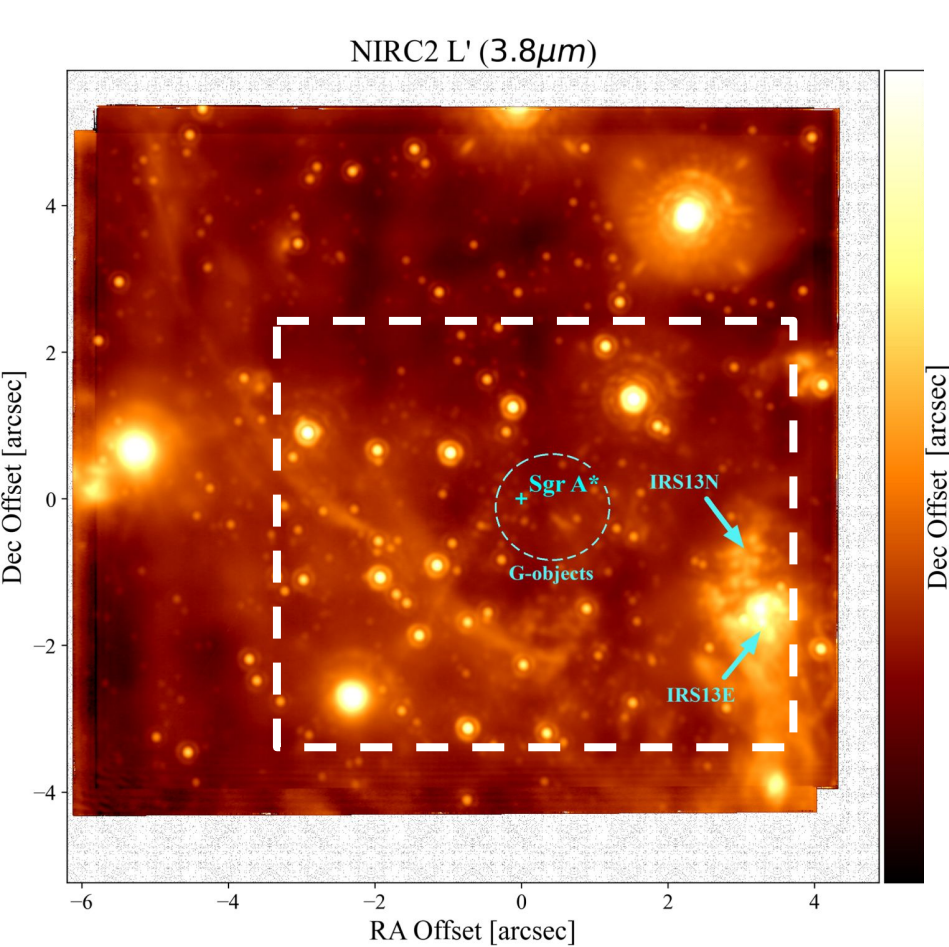
2010-2020: More simultaneous multi-wavelength flares

2025: JWST MIR detection with synchrotron cooling evidence

The Galactic Center

$d=8.0$ kpc





At the GC, 1 arcsecond = 0.04 parsec = 8000 AU

What do we expect to 'see'?



Sgr A*: *Our* Black Hole



ZOOM⁺

What do we expect to 'see'?

Stars (and clusters)

- Seen well in UV, optical, IR
- Orbits used to measure BH mass

Gas and dust

- LOTS of it, eventually feeds the BH
- Also blocks all of UV/optical light, making it hard to see anything :(

The Central Region

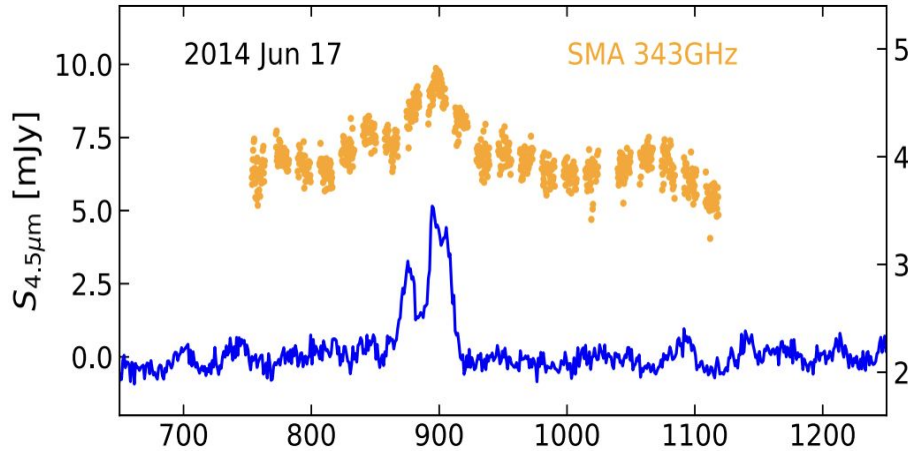
The Black Hole itself

- Cannot see BH, only accretion disk
- So small it could only be seen in radio with a telescope the size of Earth

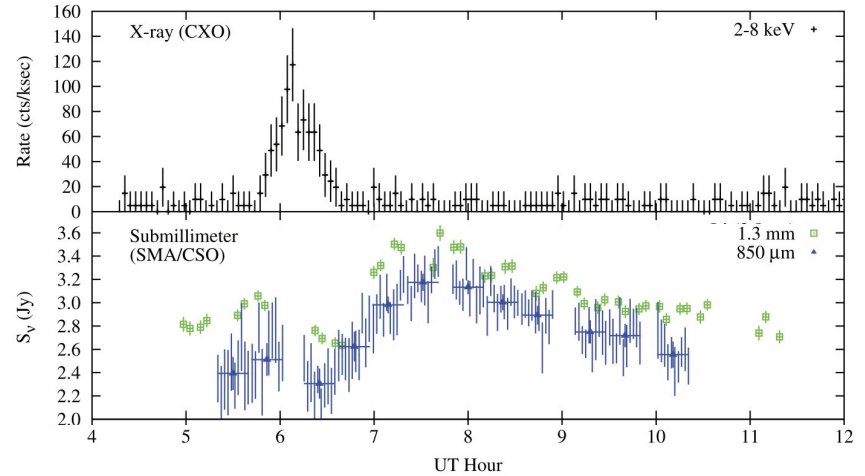
'Flares' in the accretion disk

- Seen as sudden spikes of brightness when a lot of material falls in
- Easier to see as change in brightness, not absolute brightness

Focus on the Flares!



Witzel+2021



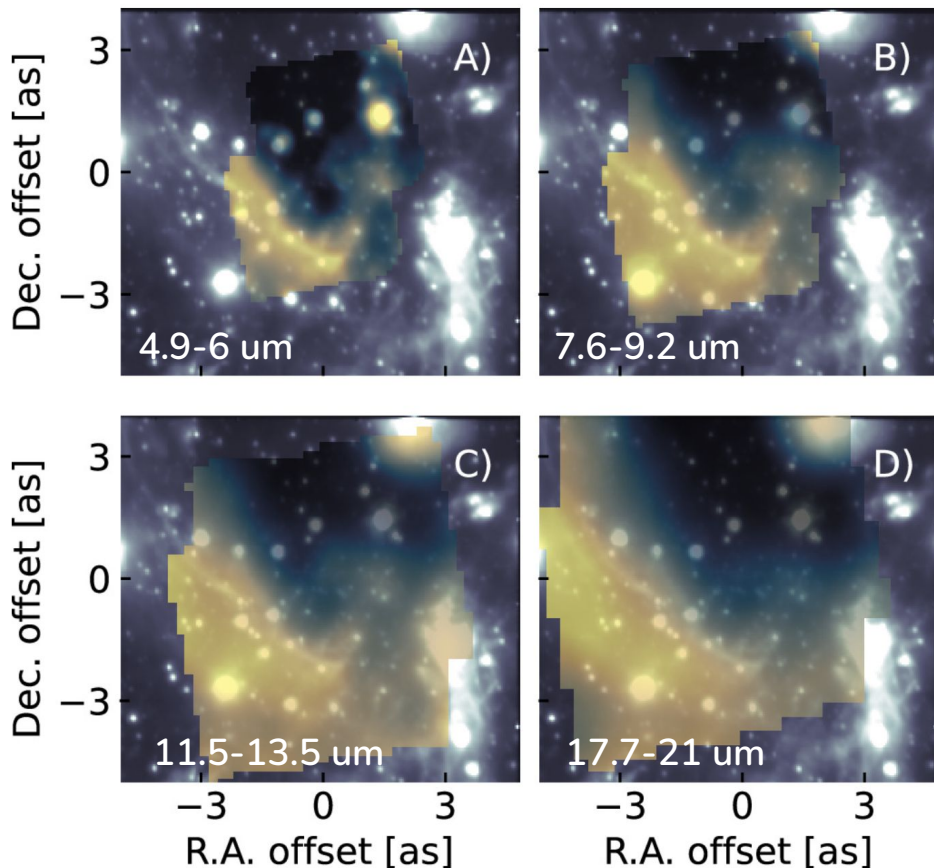
Marrone+2008

Sgr A* and its flares/variability have been seen in **radio, near-IR and X-ray** , but not in mid-IR:

flaring mechanism still not well-understood !

JWST: Window to mid-IR (5–28 μm)!

Data Overview



Sgr A* and its flares/variability have been seen in **radio, NIR and X-ray**, but not in mid-IR
JWST: Window to MIR!

Broad Data Description :

- Spatial coverage: very small (okay because the BH is like a point)
- Minute-scale time resolution over 30 hours : crucial to track changes in brightness (i.e. flares)
- Wavelength resolution : helps understand the cause of flare better!

Data Overview and Scales

First detection of a mid-IR flare from Sgr A*

Helped us understand the cause/mechanism behind flares!

Constraints on the presence of a second BH

Second BH means second source of brightness spikes. We rule that out.

Results

First high-resolution dust map of the Galactic Centre

Dust blocks light. More dust => more light lost. Need a measure of that.

High-res map of elemental abundances around BH

Is the material there similar to what makes up stars and gas in the rest of the Galaxy?

Data Overview and Scales

**First detection of a mid-IR
flare from Sgr A***

**Constraints on the presence
of a second BH**

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**First high-resolution dust
map of the Galactic Centre**

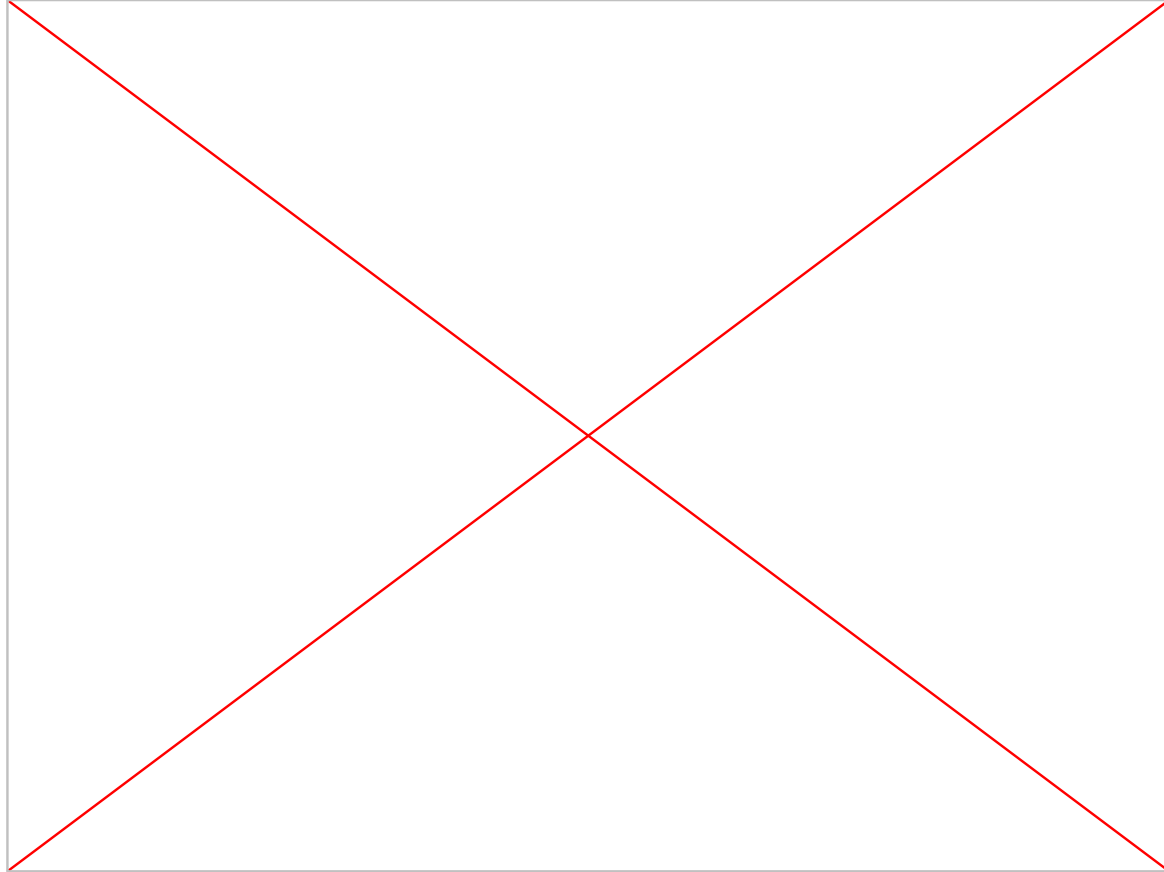
**High-res map of elemental
abundances around BH**

First detection of a mid-IR flare from Sgr A*

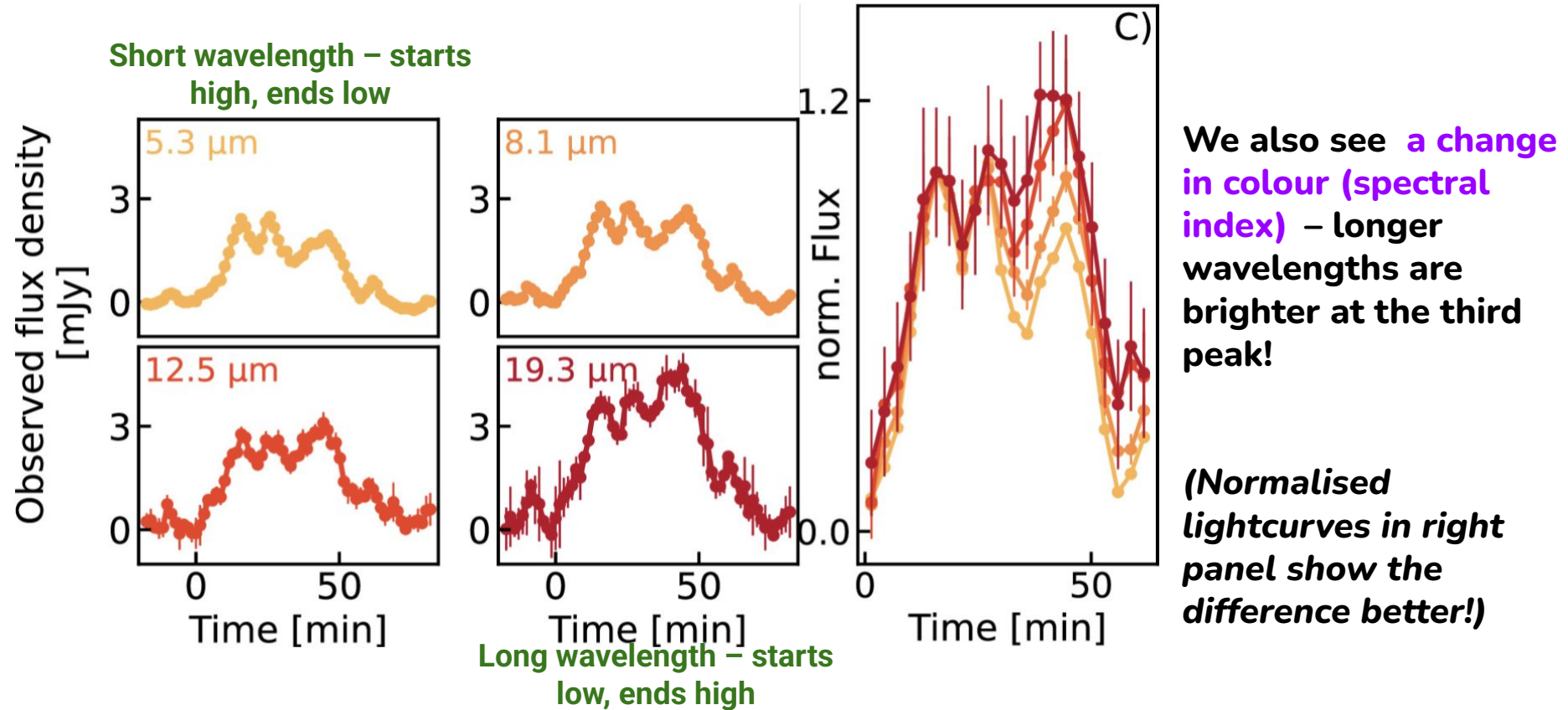
Sgr A* and its flares have been seen in **radio, NIR and X-ray**, but not in mid-IR

Flare Origin : Most likely **orbiting blob of electrons**, radiating in the tangled **magnetic field** of accretion disk

[video on right – multiple peaks due to Doppler shift in orbit around BH]

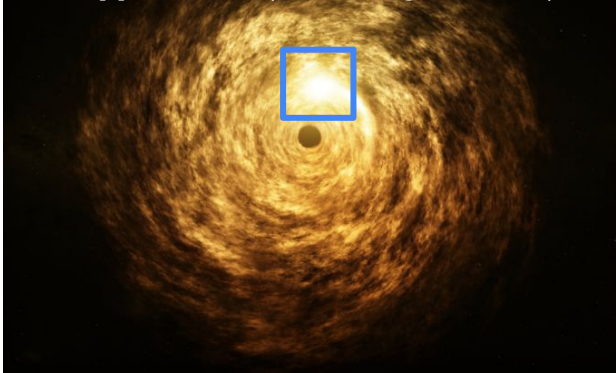


First detection of a mid-IR flare from Sgr A*

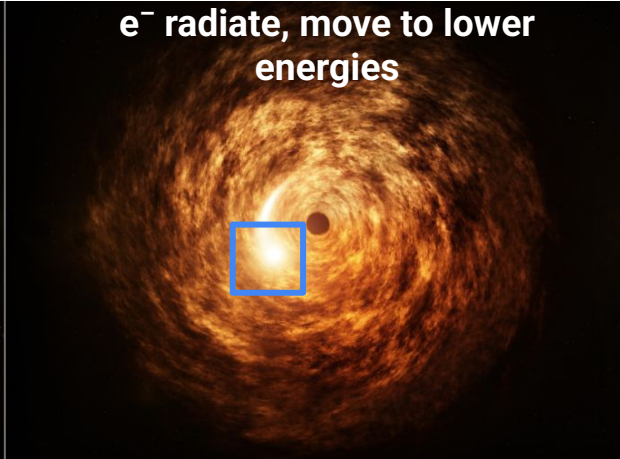


First detection of a mid-IR flare from Sgr A*

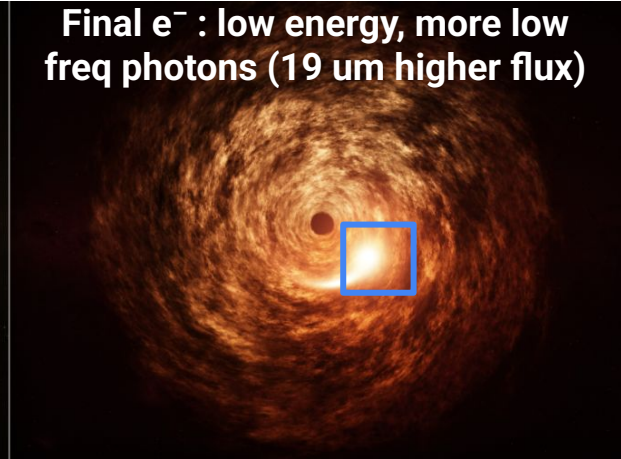
Initial e^- : high energy, more high freq photons (5 μm higher flux)



e^- radiate, move to lower energies



Final e^- : low energy, more low freq photons (19 μm higher flux)



Change in colour modelled well by **synchrotron cooling of highest energy electrons**

Why study more such flares?

1. Missing link for studies of **multiwavelength flares** and their mechanisms
2. Understanding the **physics of flare formation**
3. Probe extremely **energetic magnetic processes**

Data Overview and Scales

**First detection of a mid-IR
flare from Sgr A***

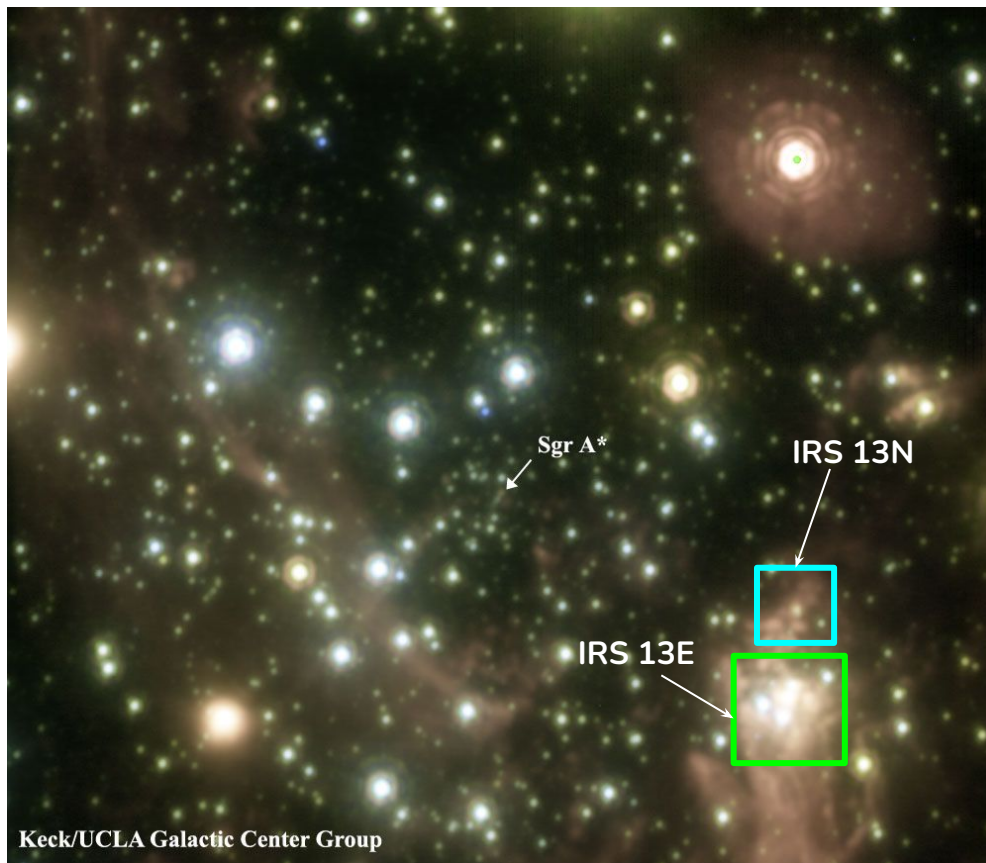
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**High-res map of elemental
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First photometric constraints on a second BH



IRS 13E – Located 3.5" (about 0.13 pc) away from Sgr A*

Hosts several **young, massive stars** that appear to form a **bound system**

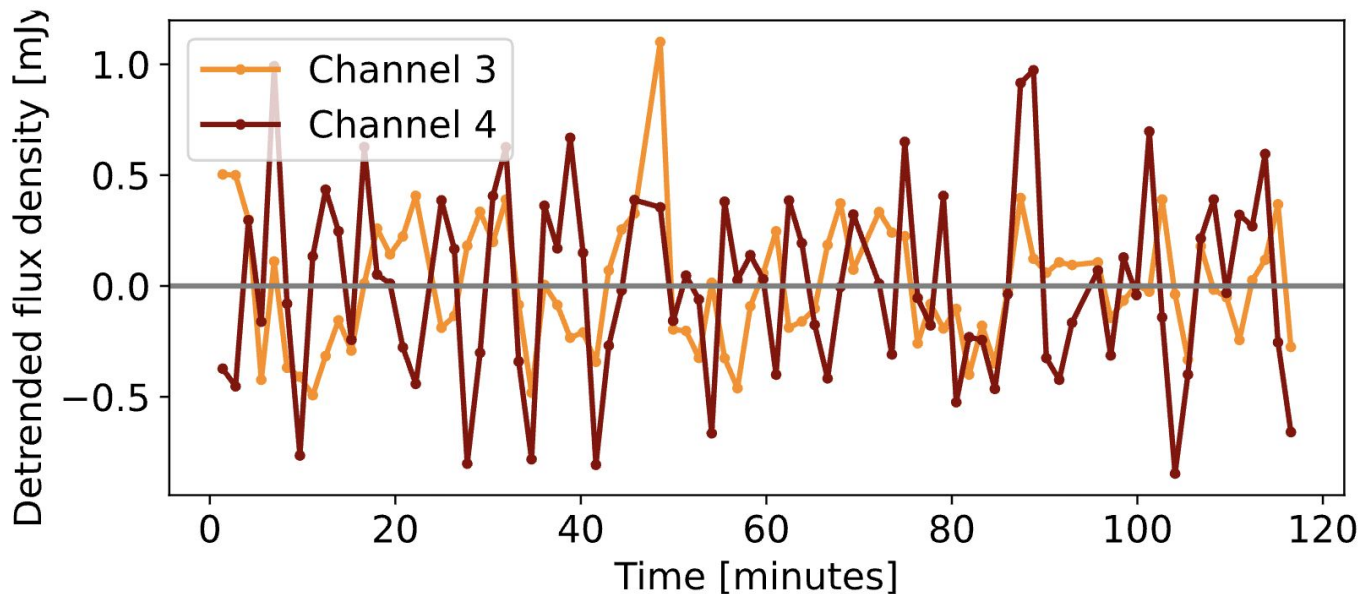
Speculation – gravitationally bound by an **intermediate mass black hole** (IMBH)

Past Constraints : Dynamical, only probe mass

Our Constraints : Photometric, probe **mass and accretion rate**

First photometric constraints on a second BH

Idea: Any BH in that region is surrounded by hot gas falling into it – must show variability (confirmed by both modelling and previous observations)

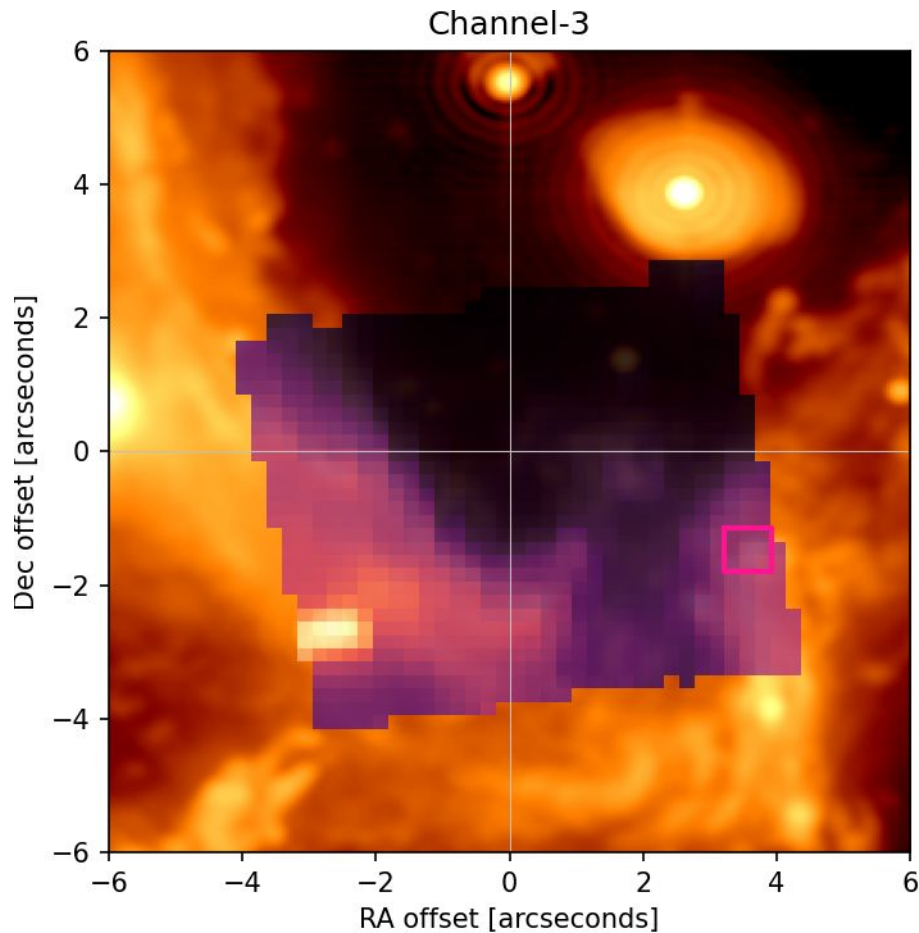


No variability seen here!

First photometric constraints on a second BH

We can get an **upper limit** on mass and accretion rate of BH — for that we need model predicted values for fluxes

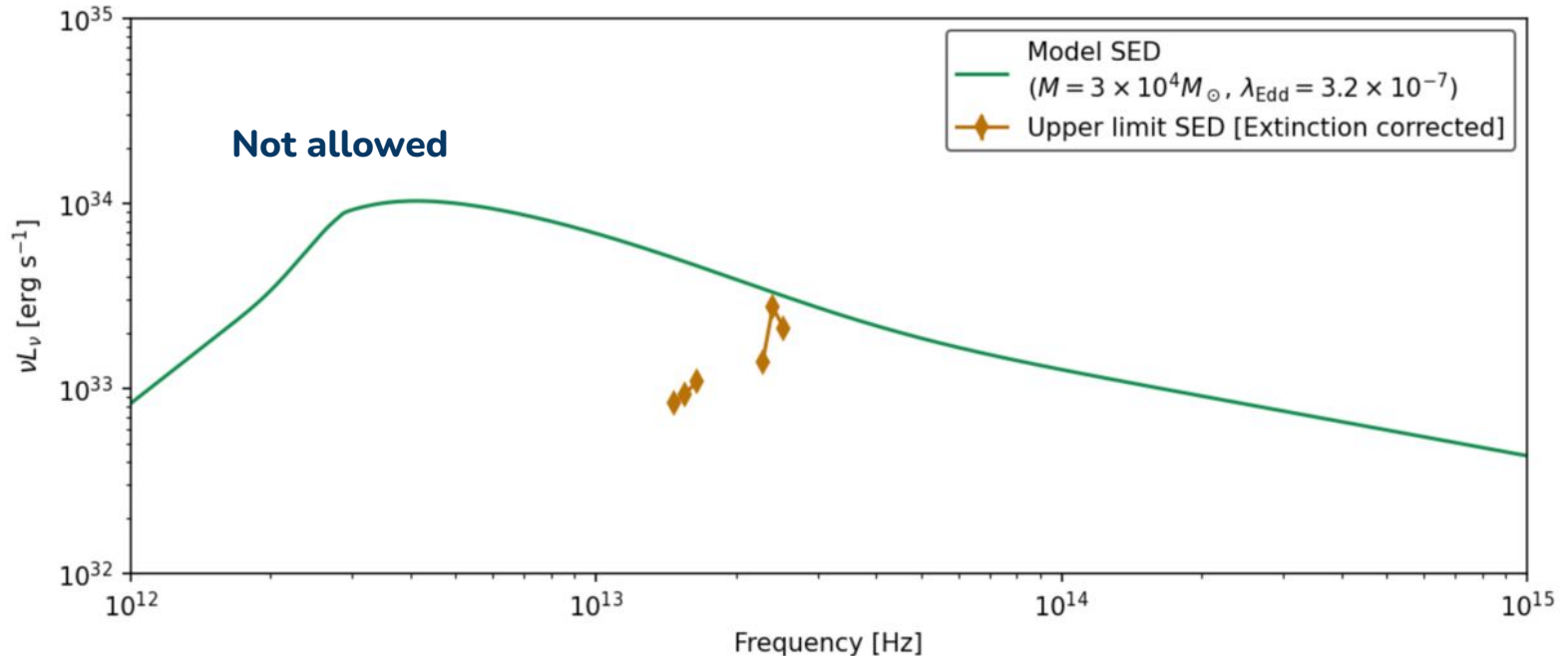
(upper limit because it is crowded, so many other objects could show some random variability too)



First photometric constraints on a second BH

Use LLAGN accretion models to calculate SED for different masses and accretion rates .

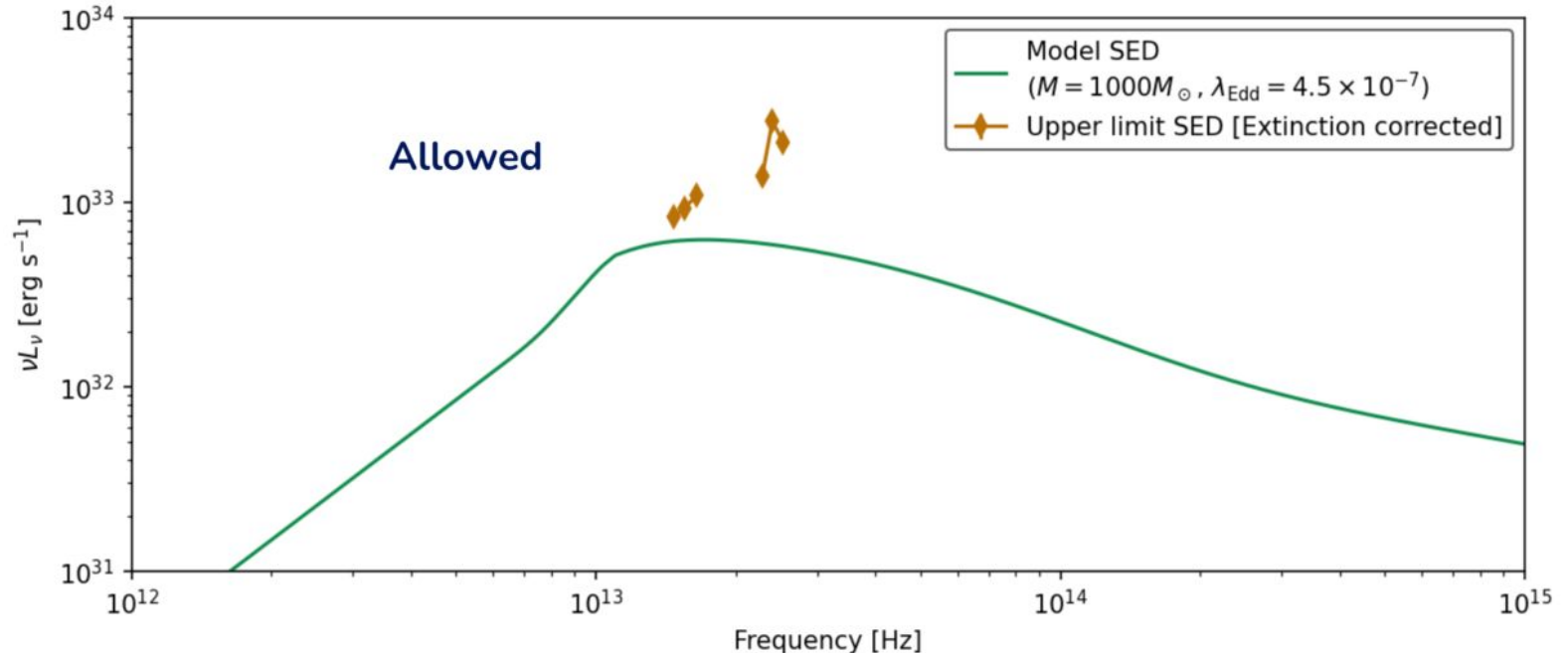
If model predicted flux > observed upper limit : parameter pair rejected as infeasible



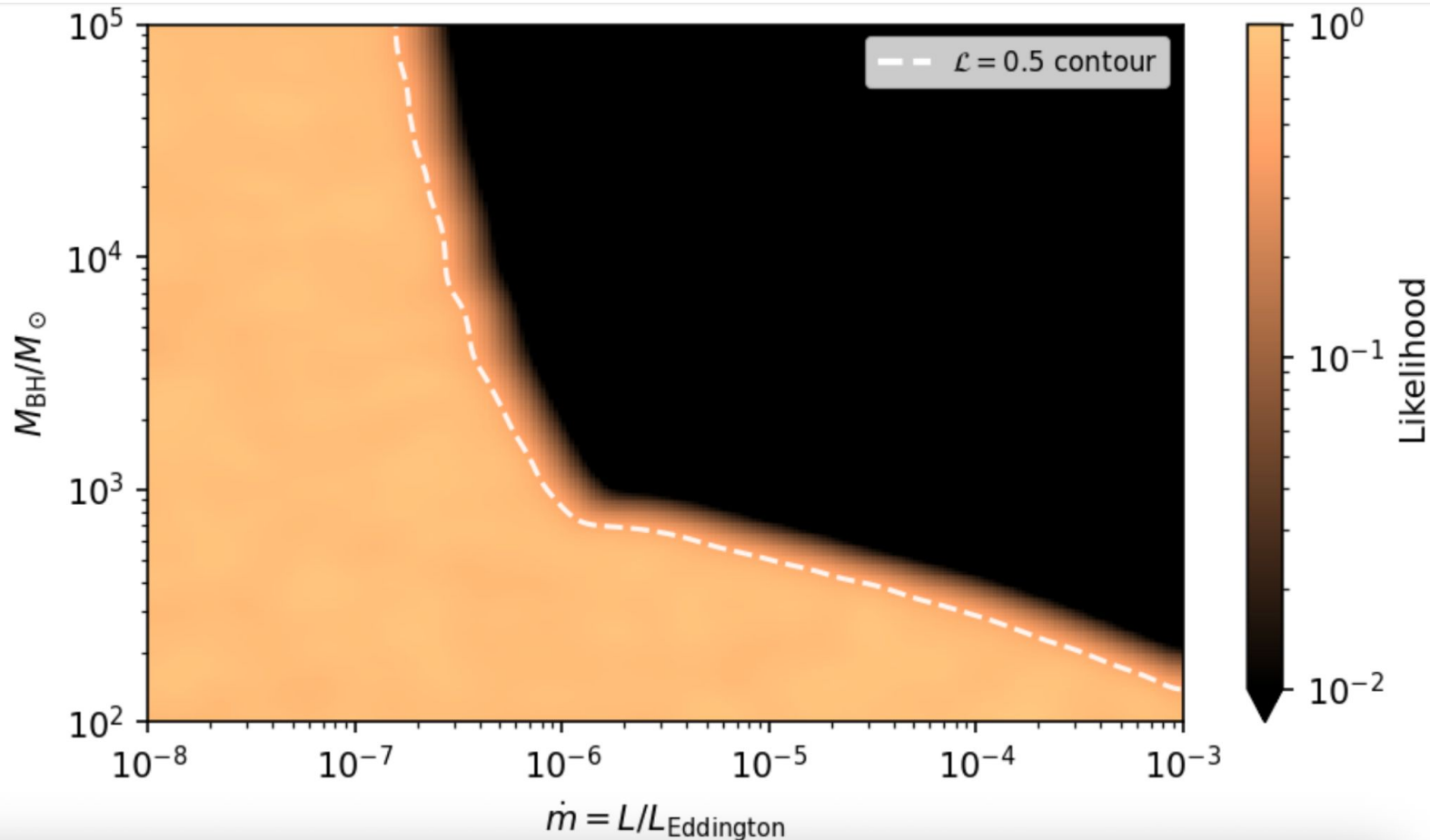
First photometric constraints on a second BH

Use LLAGN accretion models to calculate SED for different masses and accretion rates .

If model predicted flux < observed upper limit : parameter pair is infeasible



First photometric constraints on a second BH



Data Overview and Scales

**First detection of a mid-IR
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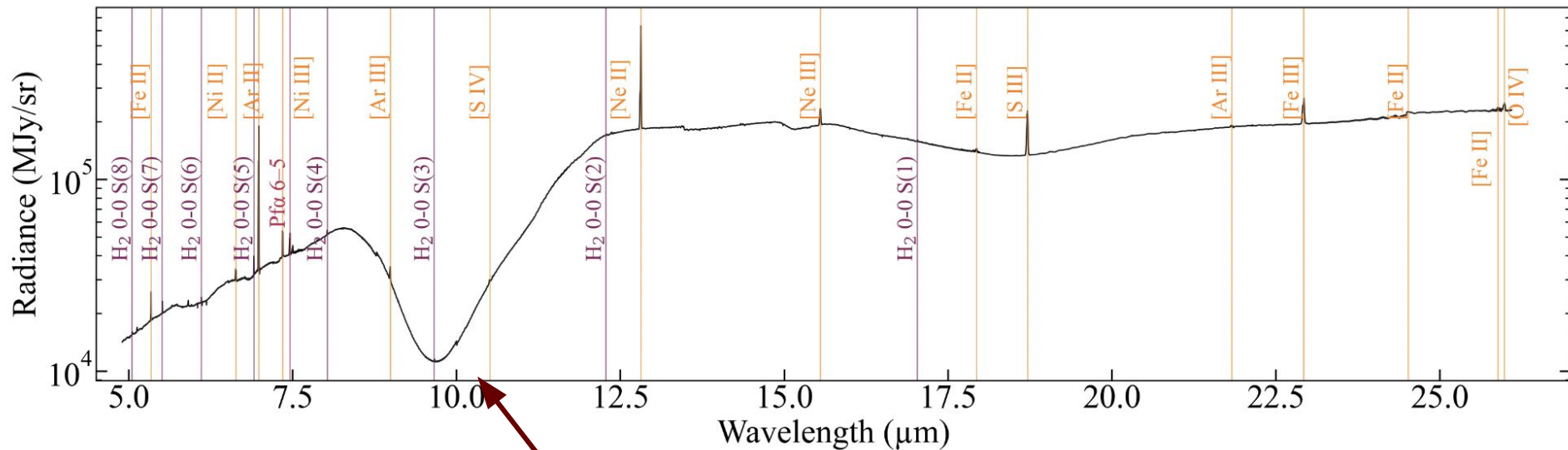
**Constraints on the presence
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**High-res map of elemental
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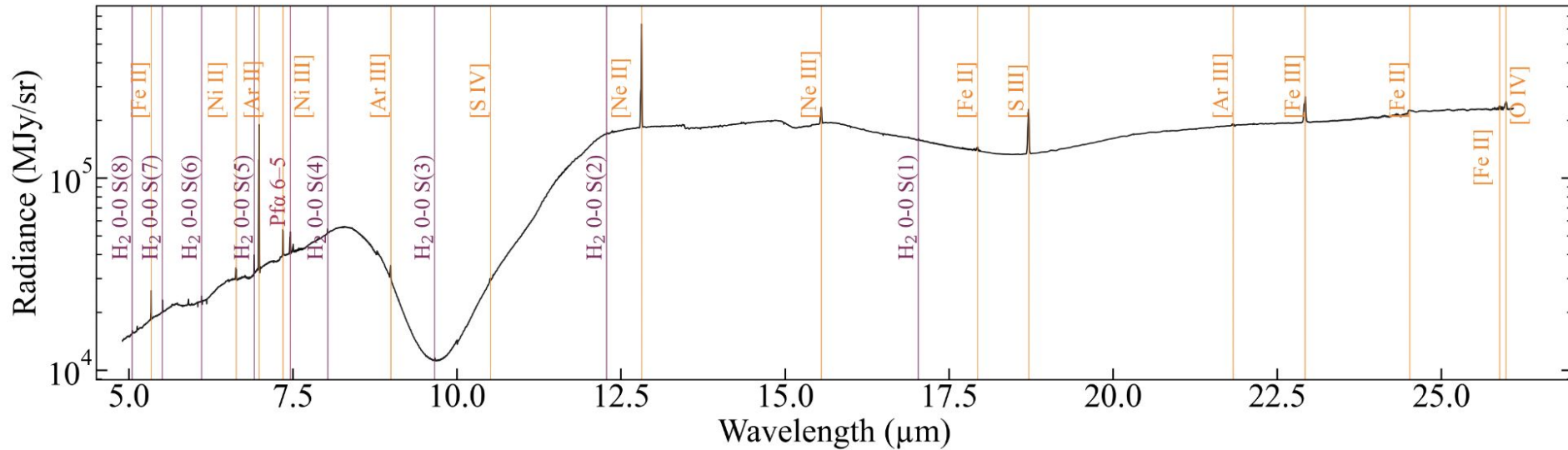
Maps of dust and elements



More extinction at
shorter wavelengths

Extra extinction by
silicates around 10
 μm

How to measure dust densities?



Spectrum has many hydrogen lines at different wavelengths

Radio observations + quantum mechanics fixes the ratio of line brightness

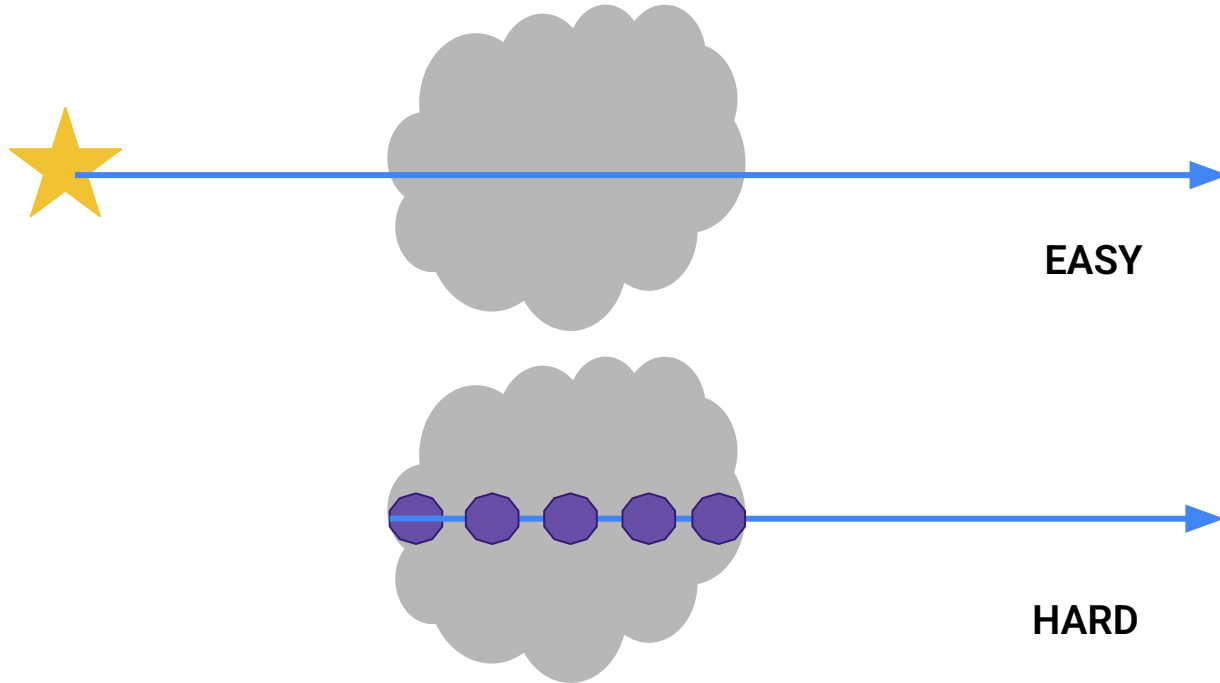
Check observed ratio of line brightness => deviations are due to dust blocking light

Measure of dust blockage (extinction) at each wavelength => Gives dust density

How to measure dust densities?

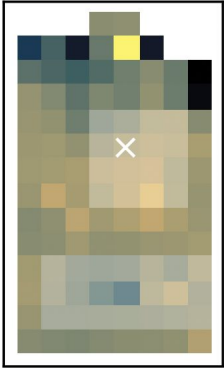
Can we do this for other wavelengths?

Problem: Most of the light is coming from gas/dust, then getting blocked by other gas/dust!

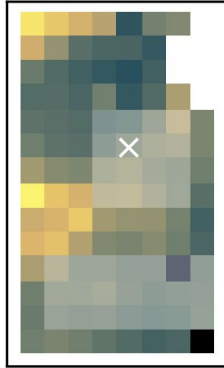


How to measure dust densities?

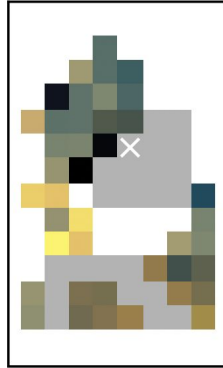
$\lambda = 5.129$



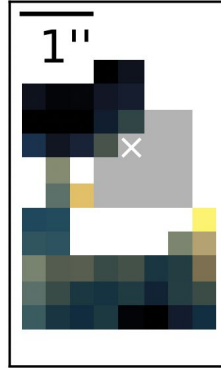
$\lambda = 5.908$



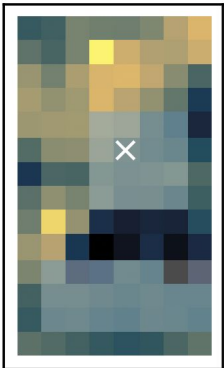
$\lambda = 5.957$



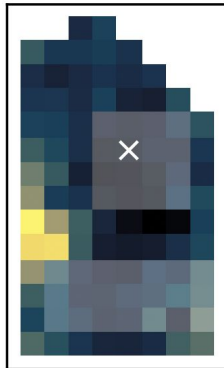
$\lambda = 6.292$



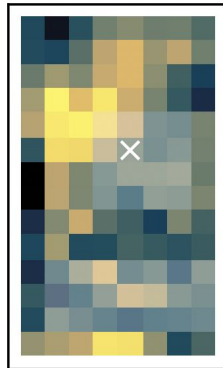
$\lambda = 11.309$



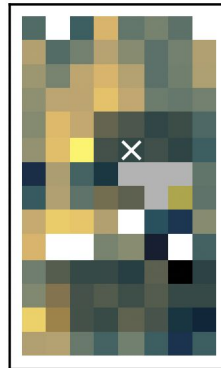
$\lambda = 12.372$



$\lambda = 16.209$



$\lambda = 19.057$



Final dust extinction maps!

Not phenomenal to look at, but:

- Allow to get **intrinsic brightness values** (by correcting for the blocked fraction of light)

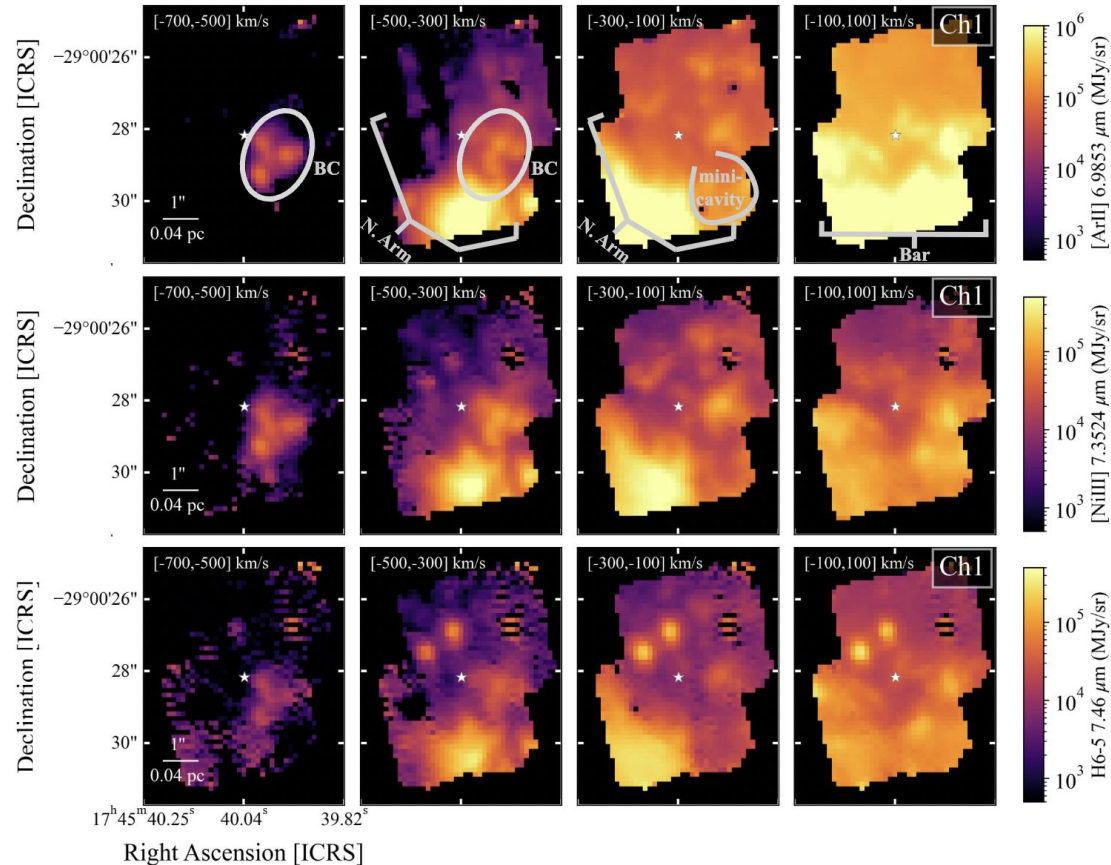
- Allow to get **intrinsic strengths of lines** for various elements in the spectrum (giving us elemental abundances)

- Dust extinction can be extrapolated to **shorter wavelengths**

How to measure elemental abundances?

- Main probe: **Line ratios** w.r.t. standard like hydrogen!
- Can also get **velocity** structures using Doppler effect

Velocity maps reveal the flow of gas going into the black hole!

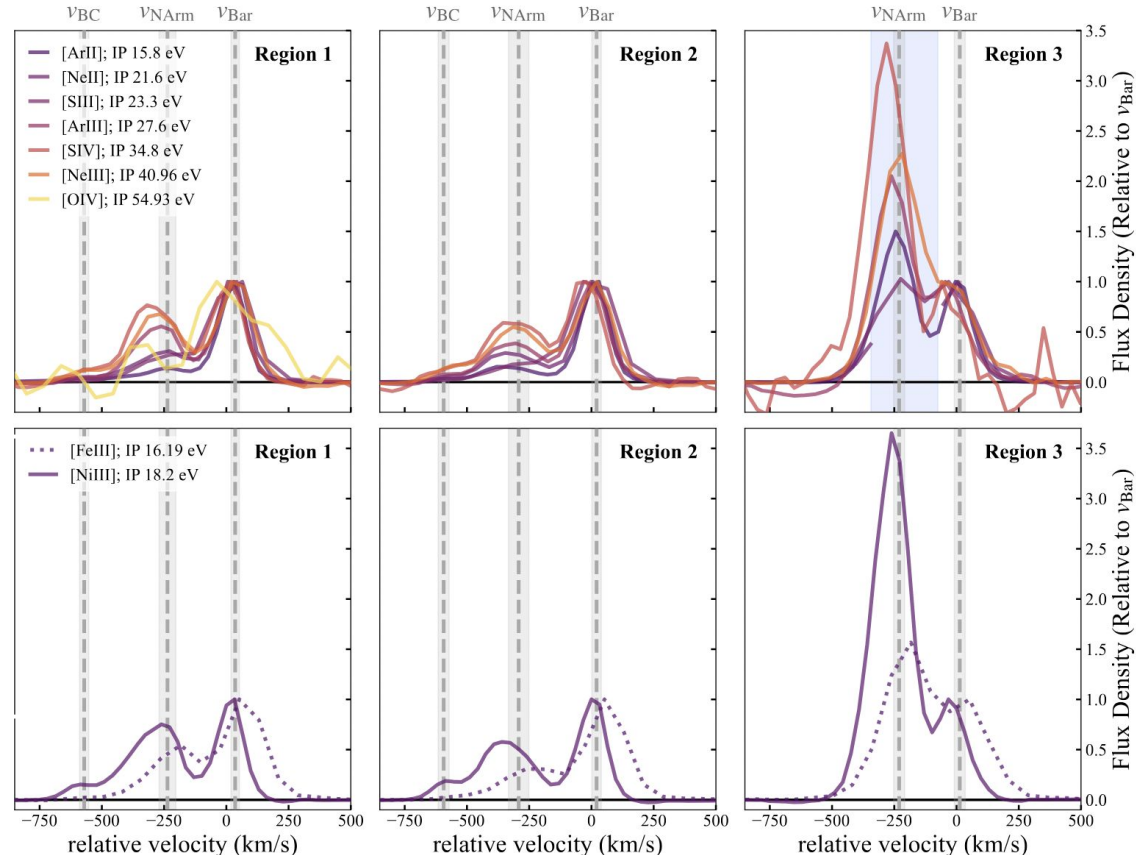


How to measure elemental abundances?

- Main probe: **Line ratios** w.r.t. standard like hydrogen!
- Can also get **velocity** structures using Doppler effect

Abundances of elements and ions is generally consistent with stars in that area

EXCEPT: Nickel is enriched more than Iron.



First detection of a mid-IR flare from Sgr A*

We now know their cause to be electrons in a magnetic field (synchrotron)

Constraints on the presence of a second BH

Large black holes with huge accretion rates are not possible

Results

First high-resolution dust map of the Galactic Centre

We have a map of dust extinction using multiple methods

High-res map of elemental abundances around BH

More nickel than iron (surprising), also recover velocity structures!

Future Work : (1) Study the full sample of flares
(2) Use the high energy species' lines to constrain UV power
(3) Simulations and theory of accretion + flares

... and much more!